BASH SCRIPTING FUNDAMENTALS

Functions

Reusable building blocks and clean script structure

Merit Advisory

Bash Scripting Fundamentals Bootcamp

Beginner-friendly lecture materials

July 2026

WHAT WE WILL COVER

Agenda

1. Section 1: Function Foundations — 4 slides
2. Section 2: Function Parameters — 4 slides
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1 Section 1

Function Foundations

1.1 Why functions · Name parentheses braces form

- Functions name a reusable piece of shell logic.
- They reduce repeated code in longer scripts.
- They create small units that are easier to test.
- The name() { } form is common Bash function syntax.
- A space is required before the opening brace body.
- Commands inside the braces run when the function is called.

Examples

```
# Repeated action
say_hi() { echo "hi $1"; }
say_hi Ada
# Named step
build() { make all; }
build
# Define
hello() { echo hello; }
# Define and call
hello() { echo hello; }
hello
```

1.2 Function keyword form · Calling functions

- Bash also accepts the function keyword.
- It is less portable but common in Bash-only scripts.
- Pick one style and use it consistently.
- Call a function by writing its name like a command.
- Arguments follow the name separated by spaces.
- The function must be defined before the call runs.

Examples

```
# Keyword
function hello { echo hello; }
# Keyword call
function build { echo build; }
build
# Simple call
greet() { echo hi; }
greet
# Call with arg
greet() { echo "hi $1"; }
greet Sam
```

1.3 Declare before use · Function call status

- Bash executes scripts from top to bottom.
- A function definition must be seen before it is called.
- Place helper definitions near the top of scripts.
- A function returns the status of its last command by default.
- That status can drive if, &&, and ||.
- Use return when the last command is not the intended status.

Examples

```
# Before main
helper() { echo help; }
helper
# Main last
main() { helper; }
helper() { echo h; }
# Status from grep
has_root() { grep -q root /etc/passwd; }
has_root && echo yes
# Status from test
exists() { [[ -e $1 ]]; }
exists file || echo missing
```

1.4 Compose small functions

- Small functions are easier to name and reuse.
- One function should do one clear job.
- Compose larger workflows from smaller helpers.

Examples

```
# Two helpers
clean() { rm -rf out; }
build() { mkdir out; }
# Workflow
run_all() { clean && build; }
run_all
```

2 Section 2

Function Parameters

2.1 First argument inside a function · All arguments with dollar at

- $1 inside a function is the function first argument.
- It is separate from the script first argument.
- Quote it when using it as data.
- "$@" expands to all function arguments safely.
- Each original argument stays a separate word.
- Use it when forwarding arguments to another command.

Examples

```
# Print first
show() { echo "$1"; }
show alpha
# Use path
size() { wc -c < "$1"; }
size file.txt
# Forward
run() { command "$@"; }
run echo hi
# List args
show() { printf '%s\n' "$@"; }
show a "b c"
```

2.2 Argument count · Default arguments

- $# is the number of arguments passed to the function.
- Use it for arity checks.
- Check counts before reading required parameters.
- Parameter expansion can provide defaults.
- Defaults make optional parameters explicit.
- Use local variables to name defaulted values.

Examples

```
# Require one
need_one() { [[ $# -eq 1 ]] || return 2; }
# Print count
count_args() { echo "$#"; }
count_args a b
# Default env
deploy() { local env=${1:-dev}; echo "$env"; }
# Default file
read_cfg() { local cfg=${1:-config.env}; echo "$cfg"; }
```

2.3 Shift inside functions · Validate function arguments

- shift removes the first function argument.
- It is useful after parsing an option or command.
- It does not shift the script arguments outside the function.
- Functions should check the inputs they require.
- Return a nonzero status when validation fails.
- Keep validation close to the top of the function.

Examples

```
# Drop command
run_cmd() { local cmd=$1; shift; "$cmd" "$@"; }
# Parse first
show_rest() { local first=$1; shift; printf '%s\n' "$@"
# Require file
read_file() { [[ -r $1 ]] || return 2; wc -l < "$1"; }
# Require int
twice() { [[ $1 =~ ^[0-9]+$ ]] || return 2; echo $(($1
```

2.4 Usage function

- A usage function prints the valid command form.
- Keep it short enough to read after an error.
- Call it from argument validation branches.

Examples

```
# Usage text
usage() { echo "usage: app FILE"; }
# Use usage
[[ $# -eq 1 ]] || { usage; exit 2; }
```

3 Section 3

Scope And Local State

3.1 Local variables · Missing local pitfall

- local creates a variable scoped to the current function.
- It prevents accidental changes to global variables.
- Use local for temporary function state.
- Assignments without local can change global variables.
- This can create surprising behavior between functions.
- Declare scratch variables local by default.

Examples

```
# Local name
greet() { local name=$1; echo "hi $name"; }
# Local count
count() { local n=0; ((n++)); echo $n; }
# Global leak
tmp=old
set_tmp() { tmp=new; }
# Fixed
set_tmp() { local tmp=new; echo "$tmp"; }
```

3.2 Readonly function variables · Global variables are shared state

- readonly prevents reassignment after a value is set.
- Inside functions, combine local and readonly for constants.
- Use it for values that must not change within a helper.
- Global variables are visible inside functions.
- They couple helpers to outer script state.
- Prefer parameters and outputs for most data flow.

Examples

```
# Local readonly
show() { local -r path=$1; echo "$path"; }
# Readonly global
readonly APP_NAME=demo
# Read global
prefix=app
name() { echo "$prefix-$1"; }
# Avoid write
set_env() { local env=$1; echo "$env"; }
```

3.3 Local arrays · Nameref intro

- Arrays can be local to a function.
- Use local -a for indexed arrays.
- Quote array expansions when passing values along.
- local -n creates a nameref to another variable.
- It lets a function update a caller variable by name.
- Use it carefully because it increases coupling.

Examples

```
# Local array
list() { local -a xs=(a b); printf '%s\n' "${xs[@]}"; }
# Build array
make_args() { local -a args=(--name "$1"); printf '%s\n
# Set by name
set_value() { local -n ref=$1; ref=$2; }
set_value result ok
# Array by name
first_item() { local -n arr=$1; echo "${arr[0]}"; }
```

3.4 Unset temporary names

- unset removes a variable name from the current scope.
- Local variables disappear automatically when a function returns.
- Use unset when a long function must clear sensitive values early.

Examples

```
# Unset local
demo() { local token=$1; unset token; }
# Unset global
cache=value
unset cache
```

4 Section 4

Return Values And Output

4.1 Return status · Return status range

- return sets the function exit status.
- A status of 0 means success.
- Use nonzero statuses for validation or command failures.
- Shell return statuses are small integers.
- Do not use return for strings or structured data.
- Use stdout for data a caller should capture.

Examples

```
# Return success
ok() { return 0; }
ok && echo yes
# Return failure
fail() { return 1; }
fail || echo no
# Bad data return
get_name() { echo Ada; }
# Status only
is_file() { [[ -f $1 ]]; }
```

4.2 Return versus echo · Capture command substitution

- return communicates success or failure.
- echo or printf communicates textual output.
- Keep status and data roles separate.
- Command substitution captures stdout from a function.
- Trailing newlines are removed by the substitution.
- Quote the assignment result when using it later.

Examples

```
# Predicate
is_prod() { [[ $1 == prod ]]; }
# Producer
make_name() { printf 'app-%s\n' "$1"; }
# Capture
now() { date +%s; }
ts=$(now)
# Capture name
slug() { echo "app-$1"; }
name=$(slug api)
```

4.3 Stdout versus stderr · Quiet predicate functions

- Stdout is for data a caller may capture.
- Stderr is for diagnostics and errors.
- Send messages to stderr when stdout must stay parseable.
- Predicate functions often print nothing.
- Their status is the answer.
- This makes them easy to use in if statements.

Examples

```
# Error output
warn() { echo "warn: $*" >&2; }
# Data output
value() { printf '%s\n' "$1"; }
# Has command
has_cmd() { command -v "$1" >/dev/null 2>&1; }
# Readable
can_read() { [[ -r $1 ]]; }
```

4.4 Multiple values with echo

- A function can print multiple lines as output.
- The caller must parse that output deliberately.
- For complex data, prefer clear formats or separate helpers.

Examples

```
# Two lines
pair() { printf '%s\n' name Ada; }
# Parse simple
vals=$(pair)
printf '%s\n' "$vals"
```

5 Section 5

Main Pattern And Libraries

5.1 Main function pattern · Pass arguments to main

- A main function names the script entry point.
- It keeps top-level code short.
- It makes script flow easier to scan.
- Use "$@" when calling main from top level.
- This preserves the original script arguments.
- Main can then validate and dispatch them.

Examples

```
# Main
main() { echo run; }
main "$@"
# Main status
main() { return 0; }
main "$@"
# Forward all
main() { printf '%s\n' "$@"; }
main "$@"
# Use first
main() { echo "cmd=$1"; }
main "$@"
```

5.2 Sourcing a library · Guard against re source

- source reads another shell file into the current shell.
- Functions defined there become available to the script.
- Use relative paths carefully when sourcing libraries.
- A guard variable can prevent loading a library twice.
- Set the guard after the first successful load.
- This protects repeated readonly declarations.

Examples

```
# Source lib
source ./lib.sh
helper
# Dot form
. ./lib.sh
helper
# Library guard
[[ ${LIB_LOADED:-} ]] && return 0
LIB_LOADED=1
# Readonly guard
[[ ${UTILS_LOADED:-} ]] && return 0
readonly UTILS_LOADED=1
```

5.3 Function library layout · Helpers before main

- Group related helper functions in a small library file.
- Keep script entry logic out of reusable libraries.
- Use clear prefixes when many libraries are sourced.
- Place helper definitions before main calls them.
- This matches Bash top to bottom execution.
- The main call should usually be near the end.

Examples

```
# Utils file
log_info() { echo "info: $*"; }
# Script use
source ./utils.sh
log_info start
# Order
helper() { echo h; }
main() { helper; }
# Call last
main() { echo run; }
main "$@"
```

5.4 Run only when executed

- BASH_SOURCE can distinguish source from execute.
- This allows a file to be both script and library.
- Call main only when the file is executed directly.

Examples

```
# Entry guard
if [[ ${BASH_SOURCE[0]} == $0 ]]; then main "$@"; fi
# Source safe
main() { echo run; }
[[ ${BASH_SOURCE[0]} == $0 ]] && main "$@"
```

6 Section 6

Error Logging And Assertions

6.1 Die helper · Error helper

- A die helper prints an error and exits.
- It centralizes fatal error formatting.
- Use it only where exiting the script is intended.
- An error helper prints to stderr without exiting.
- It lets callers decide whether to return or continue.
- This is useful inside reusable functions.

Examples

```
# Die
die() { echo "error: $*" >&2; exit 1; }
# Use die
[[ -r $cfg ]] || die "missing config"
# Error
error() { echo "error: $*" >&2; }
# Return after error
need_file() { [[ -f $1 ]] || { error missing; return 2
```

6.2 Info logging helper · Warn logging helper

- An info helper marks normal progress messages.
- Send logs to stderr when stdout is data.
- Keep the format consistent across scripts.
- A warn helper marks nonfatal problems.
- It should not exit on its own.
- Use it when the script can continue safely.

Examples

```
# Info
info() { echo "info: $*" >&2; }
# Use info
info "starting build"
# Warn
warn() { echo "warn: $*" >&2; }
# Use warn
[[ -f optional.env ]] || warn "no optional env"
```

6.3 Assert command helper · Assert file helper

- An assert helper checks an invariant during a script.
- Failed assertions should return or exit clearly.
- Use assertions for programmer expectations, not user input.
- A file assertion names a common required condition.
- It keeps call sites short and readable.
- Return nonzero rather than exiting inside reusable libraries.

Examples

```
# Assert
assert() { "$@" || die "assert failed: $*"; }
# Assert use
assert test -d src
# Require file
require_file() { [[ -f $1 ]] || return 2; }
# Use require
require_file config.env || die "need config"
```

6.4 Command wrapper

- A wrapper function adds policy around another command.
- It can add logging, defaults, or validation.
- Forward arguments with "$@" to preserve words.

Examples

```
# Git wrapper
git_checked() { git "$@" || die "git failed"; }
# Curl wrapper
fetch() { curl -fsS "$@"; }
```

7 Section 7

Advanced Function Mechanics

7.1 Functions and exit · Set e inside functions

- exit inside a function exits the shell or script.
- return exits only the function.
- Library functions should usually return instead of exit.
- set -e can interact with function calls in surprising ways.
- Commands tested by if or || are treated specially.
- Use explicit status handling for important failures.

Examples

```
# Return
check() { [[ -e $1 ]] || return 2; }
# Exit fatal
fatal() { echo bad >&2; exit 1; }
# Explicit return
copy() { cp "$1" "$2" || return; }
# If tested
if risky_fn; then echo ok; fi
```

7.2 Err trap caution · Recursion caution

- ERR traps can run when commands fail under set -e.
- Functions make trap flow harder to reason about.
- Keep trap based error handling small and tested.
- Bash functions can call themselves recursively.
- Deep recursion is usually a poor fit for shell scripts.
- Prefer loops for repeated shell work.

Examples

```
# Err trap
trap 'echo failed >&2' ERR
set -E
# Function failure
work() { false; }
work
# Recursive shape
countdown() { (( $1 == 0 )) || countdown $(($1-1)); }
# Loop alternative
for ((n=3; n>=0; n--)); do echo $n; done
```

7.3 Overriding commands · Builtin command bypass

- A function can have the same name as a command.
- This changes what unqualified calls run.
- Use command to bypass a function wrapper.
- The command builtin skips shell function lookup.
- It is useful inside wrappers that call the original command.
- It can also check external command availability.

Examples

```
# Wrapper
ls() { command ls --color=auto "$@"; }
# Bypass
command ls
# Call original
grep() { command grep --color=auto "$@"; }
# Find command
command -v bash >/dev/null
```

7.4 Unset function

- unset -f removes a function definition.
- It is useful in interactive shells and tests.
- Avoid removing shared helpers during normal script flow.

Examples

```
# Remove function
tmp_fn() { echo tmp; }
unset -f tmp_fn
# Replace wrapper
unset -f ls
```

8 Section 8

Function Introspection

8.1 Declare f · Type t function

- declare -f prints function definitions.
- It is useful for debugging sourced libraries.
- Use a function name to inspect one definition.
- type -t reports what kind of name Bash resolves.
- It prints function for function names.
- It is useful before overriding or dispatching names.

Examples

```
# Show function
hello() { echo hi; }
declare -f hello
# All functions
declare -f
# Check type
hello() { :; }
type -t hello
# Command type
type -t cd
```

8.2 Command V · Declare F

- command -V describes how a name will be interpreted.
- It can show functions, builtins, aliases, and files.
- Use it when debugging resolution problems.
- declare -F lists function names without bodies.
- It is shorter than declare -f for inventories.
- Use it to confirm a library loaded expected helpers.

Examples

```
# Describe
hello() { :; }
command -V hello
# Describe ls
command -V ls
# List names
declare -F
# Has name
declare -F helper >/dev/null && echo loaded
```

8.3 Exported functions caution · Readonly functions

- export -f can pass functions to child Bash processes.
- It is Bash specific and should be used sparingly.
- Prefer script files for reusable child process behavior.
- readonly -f prevents a function from being redefined or unset.
- It can protect critical helpers in controlled scripts.
- Use it sparingly because it makes tests harder.

Examples

```
# Export function
helper() { echo h; }
export -f helper
# Child bash
bash -c helper
# Protect
die() { exit 1; }
readonly -f die
# Check readonly
readonly -f
```

8.4 Namespace prefixes

- Prefixes reduce function name collisions.
- They are helpful in sourced libraries.
- Use consistent names such as log_info or path_join.

Examples

```
# Prefixed log
log_info() { echo "info: $*"; }
# Prefixed path
path_join() { echo "$1/$2"; }
```

9 Section 9

Manual Testing And Documentation

9.1 Manual function calls · Interactive source testing

- Source a file and call functions directly while developing.
- This shortens the edit and test loop.
- Use simple arguments that cover success and failure.
- An interactive shell can load your helpers with source.
- declare -f confirms the definitions are present.
- unset -f can clean up test definitions.

Examples

```
# Source then call
source ./script.sh
my_fn test
# Call failure
my_fn missing || echo failed
# Load
source ./lib.sh
declare -F
# Cleanup
unset -f helper
```

9.2 Temporary argument tests · Capture output tests

- Small test calls can exercise argument handling.
- Include spaces in sample arguments.
- Verify both output and status.
- Command substitution can check function output.
- Compare captured output to the expected value.
- Quote variables in test comparisons.

Examples

```
# Space arg
show() { printf '<%s>\n' "$@"; }
show "a b"
# Status arg
need_one() { [[ $# -eq 1 ]]; }
need_one a b || echo bad
# Output check
got=$(slug api)
[[ $got == app-api ]]
# Print failure
[[ $got == ok ]] || echo "got $got"
```

9.3 Compare status tests · ShellCheck SC2155 note

- Status tests check whether predicates succeed or fail.
- Use if or && and || for manual checks.
- Avoid reading $? when direct conditionals are clearer.
- SC2155 warns about declaring and assigning in one command.
- Separate declaration from assignment when status matters.
- This is especially useful with command substitution.

Examples

```
# Predicate pass
is_num() { [[ $1 =~ ^[0-9]+$ ]]; }
is_num 42 && echo pass
# Predicate fail
is_num x || echo fail
# Separated local
local value
value=$(cmd)
# Simple local
local name=$1
```

9.4 Documentation comments

- A short comment can describe a function contract.
- Mention inputs, output, and status when helpful.
- Keep comments close to the function definition.

Examples

```
# Contract
# Prints a slug for one name.
slug() { echo "app-$1"; }
# Status comment
# Returns 0 when the path is readable.
can_read() { [[ -r $1 ]]; }
```

10

Section 10

Refactoring Into Functions

10.1 Extract repeated code · Pure functions

- Repeated command sequences are good function candidates.
- Name the extracted behavior by its purpose.
- Keep the call site shorter after extraction.
- A pure style uses parameters and stdout without hidden state.
- It is easier to test and reuse.
- Shell functions are often not perfectly pure, but the idea helps.

Examples

```
# Extract
backup() { cp "$1" "$1.bak"; }
# Reuse
backup app.conf
backup db.conf
# Pure slug
slug() { printf '%s\n' "${1// /-}"; }
# Capture pure
name=$(slug "api server")
```

10.2 Side effect functions · Limit global dependencies

- Side effect functions change files, directories, or process state.
- Name them with verbs so the change is obvious.
- Validate inputs before making changes.
- Functions that rely on globals are harder to reuse.
- Pass paths and options as arguments instead.
- Keep unavoidable globals readonly when possible.

Examples

```
# Make directory
ensure_dir() { [[ -d $1 ]] || mkdir -p "$1"; }
# Remove file
remove_file() { [[ -f $1 ]] && rm "$1"; }
# Pass path
load_cfg() { source "$1"; }
# Avoid hidden path
run_tool() { local bin=$1; shift; "$bin" "$@"; }
```

10.3 Small predicate helpers · Wrap pipelines

- Predicate helpers wrap common tests behind clear names.
- They should return status and usually print nothing.
- This makes if statements read like intent.
- A pipeline can be hidden behind a named function.
- The name documents what the pipeline produces.
- Be clear whether stdout is data or logs.

Examples

```
# Predicate
is_readable_file() { [[ -f $1 && -r $1 ]]; }
# Use predicate
if is_readable_file config; then echo ok; fi
# Count errors
count_errors() { grep -c ERROR "$1"; }
# Top names
top_names() { cut -d, -f1 "$1" | sort; }
```

10.4 Teardown helpers

- Cleanup code is a good candidate for a helper.
- A named cleanup function can be used with trap.
- Keep cleanup safe to call more than once.

Examples

```
# Cleanup
cleanup() { rm -rf "$tmpdir"; }
# Trap cleanup
trap cleanup EXIT
```

11 Section 11

Data Flow Techniques

11.1 Printf v assignment · Nameref output

- printf -v assigns formatted text to a variable by name.
- It avoids command substitution for simple formatting.
- Use it when building strings inside functions.
- A nameref can act like an output parameter.
- The caller passes the variable name to update.
- Document this clearly because it is less obvious than stdout.

Examples

```
# Assign formatted
printf -v name 'app-%s' "$1"
# Inside function
make_name() { printf -v result 'app-%s' "$1"; }
# Output variable
make_name() { local -n out=$1; out=app; }
make_name result
# Status with output
find_cfg() { local -n out=$1; out=config.env; }
```

11.2 Global output discouraged · Arrays passed by name

- Writing results into globals couples caller and function.
- It can be acceptable in tiny scripts but scales poorly.
- Prefer stdout or nameref outputs for reusable helpers.
- Bash arrays are often passed to functions by variable name.
- local -n lets the function read or update that array.
- Avoid using the same nameref name as the caller array.

Examples

```
# Global result
RESULT=
set_result() { RESULT=ok; }
# Stdout result
get_result() { echo ok; }
# Read array
first() { local -n a=$1; echo "${a[0]}"; }
# Append array
add_item() { local -n a=$1; a+=("$2"); }
```

11.3 Read loop in function · Mapfile in function

- A function can own a while read loop.
- Pass the input path as an argument.
- Keep the loop output format stable.
- mapfile can load function input into a local array.
- Use -t to remove trailing newlines.
- This is useful when multiple passes over lines are needed.

Examples

```
# Print lines
print_file() { while IFS= read -r l; do echo "$l"; done
# Count lines
count_file() { local n=0; while read -r _; do ((n++)); 
# Local mapfile
load_lines() { local -a lines; mapfile -t lines < "$1"
# Print first
first_line() { local -a l; mapfile -t l < "$1"; echo "
```

11.4 Stable output format

- Function output becomes an interface for callers.
- Changing columns or separators can break scripts.
- Use printf for predictable formatting.

Examples

```
# Key value
emit_pair() { printf '%s=%s\n' "$1" "$2"; }
# One per line
emit_list() { printf '%s\n' "$@"; }
```

12 Section 12

Checkpoint Labs

12.1 Build usage lab · Refactor script lab

- Write a usage function for one script command.
- Call it when arguments are missing or invalid.
- Return status 2 for usage errors.
- Find two repeated command groups in a script.
- Extract each group into a named function.
- Call the new functions from main.

Examples

```
# Usage
usage() { echo "usage: app build|test"; }
# Usage error
[[ $# -gt 0 ]] || { usage; exit 2; }
# Extract build
build_app() { make build; }
# Main uses
main() { build_app && test_app; }
```

12.2 Validator lab · Logging library lab

- Create predicate functions for common input checks.
- Keep predicate output quiet.
- Use the predicates in if statements.
- Create info, warn, and error helpers.
- Send all log messages to stderr.
- Use a consistent prefix for each severity.

Examples

```
# Integer predicate
is_int() { [[ $1 =~ ^-?[0-9]+$ ]]; }
# Path predicate
is_dir() { [[ -d $1 ]]; }
# Info warn
info() { echo "info: $*" >&2; }
warn() { echo "warn: $*" >&2; }
# Error
error() { echo "error: $*" >&2; }
```

12.3 Retry function lab · Dispatch function lab

- Wrap retry behavior in a function.
- Accept the command and its arguments with "$@".
- Return failure after the final attempt.
- Use case inside main to choose a function.
- Keep command names separate from implementation details.
- Route unknown commands to usage.

Examples

```
# Retry shape
retry() { for i in 1 2 3; do "$@" && return 0; done; re
# Use retry
retry curl -fsS http://localhost
# Dispatch
case $1 in build) build ;; test) test_all ;; *) usage
# Function command
build() { echo build; }
test_all() { echo test; }
```

12.4 Function checklist

- Name the function by the job it performs.
- Declare local variables for temporary state.
- Separate status, logs, and data output deliberately.

Examples

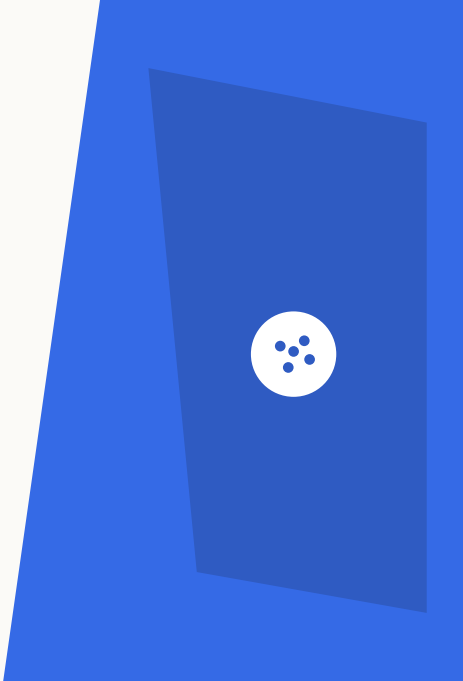
```
# Checklist helper
has_file() { [[ -f $1 ]]; }
# Checklist output
make_slug() { printf '%s\n' "${1// /-}"; }
```

NEXT STEP

Practice every example in your terminal

Then open the next module deck

Merit Advisory
04 · Functions



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